Data Cleaning and Sample Construction

Replicating the MTurk Sample of DellaVigna and Pope (2018)

Qianyu Chen | Replication note

1. Objective and data

I reconstruct the MTurk analysis sample in DellaVigna and Pope (2018) from the released `MTurkData.dta` file, following the sample restrictions in Section 3.1.2 of the paper. Starting from 12,838 records, my Python cleaning script retains 9,863 observations, compared with 9,861 in the published sample. I first wrote the cleaning rules from the paper's description rather than translating the authors' Stata code, and then used the released `Clean MTurk Data.do` file to trace the remaining differences.

2. Sample construction

Table 1 reports my sequential exclusions (Panel A) alongside the counts reported in the paper (Panel B).

Table 1: Sample selection: my script and published counts

Step	Excluded	Remaining
Panel A. My cleaning script (Python)		
Raw sample	—	12,838
Inferred Qualtrics outage interval	720	12,118
Score above 4,000	48	12,070
Not completed (`finished != 1`)	1,733	10,337
Repeated MTurk ID	133	10,204
Unapproved HIT or missing worker ID	200	10,004
Missing or non-positive effort	137	9,867
Missing or unrecognized treatment	4	9,863
Final sample		**9,863**
Panel B. Counts reported in DellaVigna and Pope (2018), Section 3.1.2		
Initial sample	—	12,838
Qualtrics technical problem	721	12,117
Score above 4,000	48	12,069
Failed to complete experiment	1,543	10,526
Stopped and logged in again	364	10,162
HIT not approved	187	9,975
No button presses	114	9,861
Final published sample		**9,861**

Notes: Panel A comes from my execution log. Panel B lists the categories and counts reported in the paper; its remaining counts are computed by subtracting these counts in sequence. Because definitions, grouping and order differ, individual rows are not directly comparable across panels. Source for Panel B: DellaVigna and Pope (2018), Section 3.1.2, p. 1042.

3. Validation against the published sample

Table 2 compares the number of observations in each of the 18 treatments with Table 3 of the paper.

Table 2: Observations by treatment: published and reconstructed

Treat.	Paper	Mine	ΔN	Treat.	Paper	Mine	ΔN
1.1	558	558	0	5.1	545	546	+1
1.2	566	567	+1	5.2	532	532	0
1.3	540	540	0	5.3	532	532	0
1.4	562	562	0	6.1	555	555	0
2.0	538	538	0	6.2	568	568	0
3.1	554	554	0	7.0	526	526	0
3.2	549	549	0	8.0	543	543	0
4.1	544	544	0	9.0	554	554	0
4.2	550	550	0	10.0	545	545	0

Sixteen counts match exactly, and treatments 1.2 and 5.1 have one extra observation each. Applying the authors' Stata rules to the raw data confirms that all 9,861 records in the published sample are also in mine, so these two records are the only difference. They amount to 0.02% of the published sample; Section 4 traces both to the outage window.

4. Understanding the discrepancies

Qualtrics outage. The paper reports 721 exclusions due to a server problem but does not give the time bounds of the outage. I locate it with a Bernoulli likelihood-ratio scan for the interval with an unusually high share of missing `buttonpresses`, using a minimum window of 20 records. The scan selects a window from 15:36:15 on May 18, 2015 to 10:54:04 on May 19, 2015, which excludes 720 records. The authors' Stata file instead drops start times from 15:34:00 on May 18 (inclusive) to 10:52:00 on May 19 (exclusive). Comparing my final sample with the records retained under the authors' rules identifies the two extra observations: one in treatment 1.2 (15:34:15) and one in treatment 5.1 (15:36:09), both on May 18. Both fall inside the authors' outage window but before the start of mine. At the other end, the records between 10:52:00 and 10:54:04 that my window excludes are all dropped at later steps under the authors' rules, so they do not affect the final sample.

Other exclusions. The intermediate counts differ mainly because of how the restrictions are grouped and ordered. In the authors' Stata code, 406 records with missing treatment and 1,137 unfinished records together account for the 1,543 non-completions reported in the paper. My script checks completion earlier and applies the duplicate-ID restriction at a different stage, leading to different exclusion counts. The order of the approval and missing-score checks also matters, as some records satisfy both conditions. Reversing these two checks reproduces the published counts without changing the final sample.

5. Reproducibility and next steps

My cleaning script, `code/01_clean.py`, writes the execution log, exclusion audit and treatment summary to `results/01_data_cleaning/` within the local `~/Desktop/replication/` folder. For the main replication and the extension, I use the authors' cleaned file `MTurkCleanedDataShort.dta`, which contains the published 9,861 worker observations. This holds the worker sample fixed when comparing the replicated results with the paper.

Reference. DellaVigna, Stefano, and Devin Pope (2018). "What Motivates Effort? Evidence and Expert Forecasts." *Review of Economic Studies* 85(2): 1029–1069.